

IB Computer Science Syllabus (2027) - Higher Level (Extension)

This list contains the additional content required **only** for Higher Level (HL) students. HL students are assessed on everything in the SL/Core list PLUS the items below.

Syllabus Point	Content Covered	Student Expected Outcome
A1.1.3	CPU vs GPU	Explain differences between CPU and GPU.
A1.1.6	Pipelining	Describe pipelining in multi-core architectures.
A1.3.5	Multitasking & Resources	Explain OS role in managing multitasking and resource allocation.
A1.3.6	Control System components	Describe the use of control system components.
A1.3.7	Control Systems applications	Explain use of control systems in real-world applications.
A1.4.1	Translation (Compilers/Interpreters)	Evaluate translation processes of interpreters and compilers.
A2.1.5	TCP/IP Model	Describe the function of the TCP/IP model.
A2.2.2	Servers	Describe the function of servers.
A2.3.4	Routing (Static/Dynamic)	Explain how static and dynamic routing move data across LANs.

A2.4.2	Network vulnerabilities	Describe common network vulnerabilities.
A2.4.3	Network countermeasures	Describe common network countermeasures.
A3.3.4	SQL Aggregate functions	Construct calculations using SQL's aggregate functions.
A3.3.5	Database Views	Describe different database views.
A3.3.6	Transactions (ACID)	Describe how transactions maintain data integrity.
A3.4.1	Alternative databases	Outline different types of databases (NoSQL, etc.).
A3.4.2	Data warehouses	Explain primary objectives of data warehouses.
A3.4.3	OLAP & Data Mining	Explain role of OLAP and data mining for business intelligence.
A3.4.4	Distributed databases	Describe features of distributed databases.
A4.2.1	Data Cleaning	Describe the significance of data cleaning.
A4.2.2	Feature Selection	Describe the role of feature selection.
A4.2.3	Dimensionality Reduction	Describe the importance of dimensionality reduction.
A4.3.1	Linear Regression	Explain how linear regression predicts continuous outcomes.
A4.3.2	Classification (Supervised)	Explain how classification

		techniques predict categorical outcomes.
A4.3.3	Hyperparameter Tuning	Explain role of hyperparameter tuning in evaluation.
A4.3.4	Clustering (Unsupervised)	Describe how clustering groups data based on similarities.
A4.3.5	Association Rules	Describe how association rules uncover relations in large data sets.
A4.3.6	Reinforcement Learning	Describe how an agent learns by interacting with its environment.
A4.3.7	Genetic Algorithms	Describe application of genetic algorithms in real-world situations.
A4.3.8	ANNs (Neural Networks)	Outline structure/function of ANNs and multi-layer networks.
A4.3.9	CNNs (Convolutional NNs)	Describe how CNNs learn spatial hierarchies in images.
A4.3.10	Model Selection	Explain importance of model selection and comparison.
B2.4.4	Recursion Concepts	Explain fundamental concept of recursion and applications.
B2.4.5	Recursive Algorithms	Construct and trace recursive algorithms.
B3.2.1	Inheritance	Explain and apply

		inheritance to promote code reusability.
B3.2.2	Polymorphism	Construct code to model polymorphism (e.g., overriding).
B3.2.3	Abstraction	Explain the concept of abstraction in OOP.
B3.2.4	Composition/Aggregation	Explain role of composition and aggregation in class relationships.
B3.2.5	Design Patterns	Explain commonly used design patterns.
B4.1.1	ADT Fundamentals	Explain properties and purpose of ADTs.
B4.1.2	Linked Lists (Theory)	Evaluate linked lists.
B4.1.3	Linked Lists (Application)	Construct and apply linked lists (singly, doubly, circular).
B4.1.4	BSTs (Binary Search Trees)	Explain structures and properties of BSTs.
B4.1.5	Sets	Construct and apply sets as an ADT.
B4.1.6	ADT Core Principles	Explain core principles of ADTs (e.g., hash tables).